

PRILL'S PROBLEM IN EVERY GENUS $g \geq 2$

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ABSTRACT. Let Y be a smooth projective connected complex curve of genus at least two. We construct a connected finite étale Galois cover $f : X \rightarrow Y$ for which every fiber divisor $f^{-1}(y)$ moves in a pencil. The essential input is Raynaud's bundle P_g , obtained by restricting the Fourier–Mukai transform of $\mathcal{O}_{\mathrm{Pic}^0(Y)}(-g\Theta)$, together with Mukai's formula showing that this transform splits as copies of one line bundle after a polarization isogeny. We choose a degree -1 twist with torsion determinant, which turns that splitting into a finite étale trivialization; the regular representation of the resulting deck group then supplies the required second section.

1. INTRODUCTION

All curves and varieties are defined over $\mathbb{C}$. Let $f : X \rightarrow Y$ be a finite morphism of smooth projective connected curves, and write $D_y = f^*y$ for its scheme-theoretic fiber over $y \in Y$. Prill asked whether

$$h^0(X, \mathcal{O}_X(D_y)) = 1$$

for a general point y whenever $g(Y) \geq 2$; see [ACGH85, Chapter VI, Exercise D]. The condition $h^0(X, \mathcal{O}_X(D_y)) \geq 2$ is closed in y by upper semicontinuity. Hence, for a fixed cover, it holds for a general point if and only if it holds for every point. Following Landesman and Litt, such a cover is called *Prill exceptional*. They constructed a finite étale Prill-exceptional cover of every complex curve of genus two and asked whether examples exist in higher genus [LL24, Theorem 1.2 and Remark 1.4].

We answer this question in every genus, and the cover we obtain is Galois.

Theorem 1.1. *Let Y be a smooth projective connected complex curve of genus $g \geq 2$. There is a connected finite étale Galois cover $f : X \rightarrow Y$ such that*

$$h^0(X, \mathcal{O}_X(f^{-1}(y))) \geq 2 \quad \text{for every } y \in Y.$$

The proof begins with Raynaud's bundle P_g on Y . This bundle has two complementary properties. First, it has no theta divisor:

$$H^0(Y, P_g \otimes L) \neq 0 \quad \text{for every } L \in \mathrm{Pic}^0(Y).$$

Second, Mukai's pullback formula says that the vector bundle on the Jacobian from which P_g is restricted becomes a direct sum of copies of a single line bundle after a finite isogeny. Thus, after a finite étale base change, the associated projective bundle is already trivial; the only remaining obstruction to trivializing the vector bundle is one scalar line bundle.

The choice of the index g is essential. Raynaud's bundle P_m has slope g/m , so P_g has slope one. We can therefore choose a line bundle B of degree -1 such that

$$E = P_g \otimes B, \quad H^0(Y, E) = 0, \quad \det E \in \mathrm{Pic}^0(Y)_{\mathrm{tors}},$$

while the Raynaud nonvanishing still gives $H^0(Y, E(y)) \neq 0$ for every y . Mukai's formula makes a finite étale pullback of E equal to $M^{\oplus g}$ for one line bundle M . The torsion determinant forces M itself to be torsion, so a further finite étale cover makes E trivial. The bundle E then comes from a representation of a finite group. One irreducible constituent has a section after twisting by every point of Y , and the same constituent occurs in the regular representation, hence in the pushforward of the structure sheaf of the final cover.

2. THE RAYNAUD–MUKAI MECHANISM

Fix a point $y_0 \in Y$, put $A = \mathrm{Pic}^0(Y)$, and let $J = \widehat{A}$ be the dual abelian variety. Write $\iota : Y \hookrightarrow J$ for the Abel–Jacobi embedding determined by y_0 . We use the theta divisor attached to the same base point,

$$\Theta = \{\alpha \in \mathrm{Pic}^0(Y) : H^1(Y, \alpha((g-1)y_0)) \neq 0\},$$

as in Hein's normalization of Raynaud's construction [Hei07, §2.1]. Let $\mathcal{P}$ be the Poincaré bundle on $J \times A$, normalized along $\{0\} \times A$ and $J \times \{0\}$, and let $p : J \times A \rightarrow A$ and $q : J \times A \rightarrow J$ be the projections.

The Fourier–Mukai transform with kernel $\mathcal{P}$ is

$$\Phi_{\mathcal{P}}(F) = Rq_*(\mathcal{P} \otimes p^*F), \quad F \in D^b(A).$$

For $\xi \in J = \text{Pic}^0(A)$, write $\mathcal{P}_{\xi} = \mathcal{P}|_{\{\xi\} \times A}$. Proper base change identifies the derived fiber at ξ with

$$L\xi^* \Phi_{\mathcal{P}}(F) \simeq R\Gamma(A, F \otimes \mathcal{P}_{\xi}).$$

Thus the transform packages, in one object on J , the cohomology of all degree-zero twists of F . Raynaud applies it to the inverse of a positive multiple of the theta line bundle.

For an ample line bundle $\mathcal{N}$ on A , the homomorphism

$$\varphi_{\mathcal{N}} : A \longrightarrow \widehat{A}, \quad x \longmapsto t_x^* \mathcal{N} \otimes \mathcal{N}^{-1},$$

measures the failure of $\mathcal{N}$ to be invariant under translation. It is an isogeny. Mukai’s formula says that this same isogeny completely splits the Fourier–Mukai transform of $\mathcal{N}^{-1}$.

Proposition 2.1 (Mukai–Raynaud). *For $m \geq 1$, set $\mathcal{N}_m = \mathcal{O}_A(m\Theta)$ and*

$$\widetilde{\mathcal{R}}_m = R^g q_*(\mathcal{P} \otimes p^* \mathcal{N}_m^{-1}), \quad \mathcal{R}_m = [-1_J]^* \widetilde{\mathcal{R}}_m, \quad P_m = \iota^* \mathcal{R}_m.$$

Then P_m is semistable and

$$\text{rk}(P_m) = m^g, \quad \deg(P_m) = gm^{g-1}, \quad H^0(Y, P_m \otimes L) \neq 0 \quad \text{for every } L \in \text{Pic}^0(Y).$$

Let $\varphi_m = \varphi_{\mathcal{N}_m} : A \rightarrow J$, and put $a_m = [-1_J] \circ \varphi_m$. Then

$$a_m^* \mathcal{R}_m \simeq \mathcal{N}_m^{\oplus m^g}.$$

Proof. For $\xi \in J$, the line bundle $\mathcal{N}_m \otimes \mathcal{P}_{\xi}^{-1}$ is ample. Since $\omega_A \simeq \mathcal{O}_A$, Serre duality and Kodaira vanishing give

$$H^i(A, \mathcal{N}_m^{-1} \otimes \mathcal{P}_{\xi}) = 0 \quad (i \neq g).$$

Riemann–Roch on the principally polarized abelian variety A gives

$$h^g(A, \mathcal{N}_m^{-1} \otimes \mathcal{P}_{\xi}) = h^0(A, \mathcal{N}_m \otimes \mathcal{P}_{\xi}^{-1}) = \chi(\mathcal{N}_m) = m^g.$$

Cohomology and base change therefore show that $\Phi_{\mathcal{P}}(\mathcal{N}_m^{-1})$ is concentrated in degree g , and that $\widetilde{\mathcal{R}}_m$ is a vector bundle of rank m^g .

Mukai’s pullback formula, applied to $\mathcal{N}_m$, is

$$\varphi_m^* \widetilde{\mathcal{R}}_m \simeq \mathcal{N}_m^{\oplus h^0(A, \mathcal{N}_m)} = \mathcal{N}_m^{\oplus m^g};$$

see [Muk81, Proposition 3.11(1)]. Raynaud records and uses exactly this formula in [Ray82, §3.1]. Since $a_m = [-1_J] \circ \varphi_m$ and $\mathcal{R}_m = [-1_J]^* \widetilde{\mathcal{R}}_m$, the displayed formula is equivalent to $a_m^* \mathcal{R}_m \simeq \mathcal{N}_m^{\oplus m^g}$.

Restricting this identity to the inverse image of the Abel–Jacobi curve shows why the resulting bundle is semistable: after a finite étale base change it is a direct sum of copies of one line bundle. Raynaud’s intersection calculation on that inverse-image curve gives $\mu(P_m) = g/m$, and hence $\deg(P_m) = gm^{g-1}$. Hein records the same construction, including the pullback by $[-1_J]$, and the resulting rank, degree, slope, and semistability in [Hei07, §2, Lemma 2.3].

It remains to explain the nonvanishing. The second Fourier–Mukai identity used by Raynaud identifies the transform of $\widetilde{\mathcal{R}}_m$ back on A with the line bundle $(-1_A)^* \mathcal{N}_m^{-1}$. Using this identity, Raynaud produces a nonzero family of sections after restricting to an Abel–Jacobi translate, and hence obtains $H^0(Y, P_m \otimes L) \neq 0$ for general $L \in \text{Pic}^0(Y)$; see [Ray82, §3.1]. Hein’s definition above is the corresponding normalized form of Raynaud’s bundle. The locus

$$\{L \in \text{Pic}^0(Y) : H^0(Y, P_m \otimes L) \neq 0\}$$

is closed by upper semicontinuity. Since it contains a dense open subset, it is all of $\text{Pic}^0(Y)$. This all-twists formulation is also recorded explicitly in [Hei07, Introduction]. $\square$

Remark 2.2 (Why the determinant is made torsion). Let $r = m^g$, and consider a twist $E = P_m \otimes B$. Pulling the isogeny a_m back along the Abel–Jacobi embedding gives a finite étale cover $\rho : Z \rightarrow Y$. Mukai’s formula then takes the form

$$\rho^* E \simeq M^{\oplus r}$$

for a line bundle M on Z . Hence

$$\mathbb{P}(\rho^* E) \simeq Z \times \mathbb{P}^{r-1}, \quad M^{\otimes r} \simeq \det(\rho^* E) = \rho^*(\det E).$$

The first identity says that the projective part of the bundle has disappeared, but it does not yet trivialize the vector bundle: the common factor M is the remaining obstruction. The determinant detects exactly

this factor. If $\det E$ is torsion, then the second identity shows that $M^{\otimes r}$, and hence M , is torsion. A finite étale cover associated with a trivialization of a power of M then makes M , and therefore E , trivial. Conversely, if a finite cover $u : U \rightarrow Z$ trivializes M , then

$$\mathrm{Nm}_u(u^*M) \simeq M^{\otimes \deg u}$$

is trivial, so M must already be torsion. Thus the torsion-determinant condition is precisely what turns Mukai's projective trivialization into a genuine finite étale trivialization of the vector bundle.

3. A DEGREE-ZERO BUNDLE WITH FINITE ÉTALE TRIVIALIZATION

We now take $m = g$ and put $r = g^g$. Proposition 2.1 gives

$$\mathrm{rk}(P_g) = \deg(P_g) = r,$$

so P_g has slope one.

Lemma 3.1. *If F is a semistable vector bundle of degree zero on Y , then*

$$V^0(F) = \{L \in \mathrm{Pic}^0(Y) : H^0(Y, F \otimes L) \neq 0\}$$

is finite.

Proof. Choose a Jordan–Hölder filtration $0 = F_0 \subset F_1 \subset \cdots \subset F_N = F$, with stable degree-zero quotients $Q_j = F_j/F_{j-1}$. If $L \in V^0(F)$, a nonzero section of $F \otimes L$ gives a nonzero map $L^{-1} \rightarrow F$. The saturation of its image is $L^{-1}(D)$ for an effective divisor D . Semistability of F gives $\deg D \leq 0$, hence $D = 0$; thus L^{-1} is a degree-zero line subbundle of F .

Choose the least j for which $L^{-1} \subset F_j$. The induced map $L^{-1} \rightarrow Q_j$ is nonzero. A nonzero morphism between stable bundles of the same slope is an isomorphism. Hence $Q_j \simeq L^{-1}$, and in particular Q_j has rank one. Thus L^{-1} is isomorphic to one of the finitely many rank-one Jordan–Hölder factors of F . $\square$

Proposition 3.2. *There is a vector bundle E on Y such that*

$$\deg E = 0, \quad H^0(Y, E) = 0, \quad \det E \in \mathrm{Pic}^0(Y)_{\mathrm{tors}}, \quad H^0(Y, E(y)) \neq 0 \quad \text{for every } y \in Y.$$

Proof. Choose $B_0 \in \mathrm{Pic}^{-1}(Y)$ and set $E_0 = P_g \otimes B_0$. Since P_g is semistable of slope one, E_0 is semistable of degree zero. Let $D = \det(E_0) \in \mathrm{Pic}^0(Y)$. Multiplication by r on the abelian variety $\mathrm{Pic}^0(Y)$ is surjective, so choose $L_0 \in \mathrm{Pic}^0(Y)$ with $L_0^{\otimes r} \simeq D^{-1}$.

The torsion subgroup of $\mathrm{Pic}^0(Y)$ is infinite, whereas $V^0(E_0)$ is finite by Lemma 3.1. We may therefore choose a torsion line bundle η such that $L_0 \otimes \eta \notin V^0(E_0)$. Set

$$B = B_0 \otimes L_0 \otimes \eta, \quad E = P_g \otimes B = E_0 \otimes L_0 \otimes \eta.$$

Then $\deg E = 0$, $H^0(Y, E) = 0$, and

$$\det E \simeq D \otimes (L_0 \otimes \eta)^{\otimes r} \simeq \eta^{\otimes r},$$

so $\det E$ is torsion. For every $y \in Y$, the line bundle $B(y) = B \otimes \mathcal{O}_Y(y)$ has degree zero. Proposition 2.1 therefore gives

$$H^0(Y, E(y)) = H^0(Y, P_g \otimes B(y)) \neq 0.$$

$\square$

Proposition 3.3. *The bundle E of Proposition 3.2 is trivialized by a connected finite étale Galois cover of Y .*

Proof. Mukai's formula has already trivialized the projective bundle associated with E ; we now remove the common line-bundle factor described in Remark 2.2. Let $a = a_g : A \rightarrow J$ and $\mathcal{N} = \mathcal{N}_g$ be as in Proposition 2.1, and form the Cartesian square

$$\begin{array}{ccc} Z & \xrightarrow{\tau} & A \\ \rho \downarrow & & \downarrow a \\ Y & \xrightarrow{\iota} & J. \end{array}$$

The morphism ρ is finite étale. Choose a connected component Z_0 of Z , and write $\rho_0 : Z_0 \rightarrow Y$ and $\tau_0 : Z_0 \rightarrow A$ for the restrictions. The image of Z_0 in Y is nonempty, open, and closed, hence ρ_0 is surjective. Proposition 2.1 gives

$$\rho_0^*E \simeq (\tau_0^*\mathcal{N} \otimes \rho_0^*B)^{\oplus r}.$$

Put $M = \tau_0^*\mathcal{N} \otimes \rho_0^*B$. Taking determinants yields

$$M^{\otimes r} \simeq \rho_0^*(\det E).$$

If $(\det E)^{\otimes d} \simeq \mathcal{O}_Y$, then $M^{\otimes rd} \simeq \mathcal{O}_{Z_0}$. Thus M is torsion.

Choose $n \geq 1$ and an isomorphism $M^{\otimes n} \simeq \mathcal{O}_{Z_0}$. Using this isomorphism to define multiplication, form the $\mathcal{O}_{Z_0}$ -algebra

$$\mathcal{A} = \mathcal{O}_{Z_0} \oplus M^{-1} \oplus \dots \oplus M^{-(n-1)}$$

and the finite cover $u : U = \operatorname{Spec}_{Z_0}(\mathcal{A}) \rightarrow Z_0$. Locally this cover has equation $t^n = v$, where v is a unit. Since the characteristic is zero, the derivative nt^{n-1} is a unit on the cover, so u is étale. The tautological root trivializes u^*M , and hence $(\rho_0 \circ u)^*E \simeq \mathcal{O}_U^{\oplus r}$.

A connected component U_0 of U still maps surjectively and finite étale to Y . Let $T \rightarrow Y$ be the normalization of Y in a Galois closure of the finite extension $\mathbb{C}(U_0)/\mathbb{C}(Y)$. Conjugates of an unramified extension of function fields are unramified, and so is their compositum. Hence $T \rightarrow Y$ is a connected finite étale Galois cover dominating $U_0 \rightarrow Y$, and the pullback of E to T is trivial. $\square$

4. THE GALOIS COVER

We fix the descent convention used below. Let $q : X \rightarrow Y$ be a connected finite étale Galois cover with group G , acting on X on the right. For a finite-dimensional left G -module V , set

$$\mathcal{E}(V) = X \times^G V = (X \times V) / ((xg, v) \sim (x, gv)).$$

Lemma 4.1. *With this convention:*

- (1) *every vector bundle on Y that becomes trivial on X is isomorphic to $\mathcal{E}(V)$ for a finite-dimensional G -module V ;*
- (2) $H^0(Y, \mathcal{E}(V)) \simeq V^G$;
- (3) $q_*\mathcal{O}_X \simeq \mathcal{E}(\operatorname{Fun}(G, \mathbb{C}))$, where $(g \cdot \varphi)(h) = \varphi(g^{-1}h)$.

Proof. Choose a trivialization $q^*F \simeq \mathcal{O}_X \otimes V$ of a vector bundle F . In this trivialization its descent isomorphisms are regular maps $X \rightarrow \operatorname{GL}(V)$. Every such map is constant because X is connected and projective whereas $\operatorname{GL}(V)$ is affine. The cocycle identity therefore gives a linear action of G on V , and the descended bundle is $\mathcal{E}(V)$.

A section of $\mathcal{E}(V)$ pulls back to a regular map $\phi : X \rightarrow V$ satisfying $\phi(xg) = g^{-1}\phi(x)$. The map ϕ is constant because X is projective and V is affine, and the equivariance condition says that its value lies in V^G . This proves the second assertion.

Étale-locally on Y , a section of the G -torsor identifies X with $Y \times G$ and $q_*\mathcal{O}_X$ with $\mathcal{O}_Y \otimes \operatorname{Fun}(G, \mathbb{C})$. Changing the local section by right multiplication by g^{-1} changes a function φ to $h \mapsto \varphi(g^{-1}h)$, which gives the stated associated bundle. $\square$

Proof of Theorem 1.1. Let E be the bundle constructed in Proposition 3.2, and choose a connected finite étale Galois cover $f : X \rightarrow Y$ that trivializes it, as supplied by Proposition 3.3. Let G be the deck group. By Lemma 4.1, $E \simeq \mathcal{E}(V)$ for a finite-dimensional complex representation V of G . Maschke's theorem gives

$$V \simeq \bigoplus_{i=1}^s V_i^{\oplus n_i}, \quad E \simeq \bigoplus_{i=1}^s W_i^{\oplus n_i}, \quad W_i = \mathcal{E}(V_i),$$

where the V_i are pairwise nonisomorphic irreducible representations. Since $H^0(Y, E) = 0$, Lemma 4.1 gives $V^G = 0$, so none of the V_i is trivial.

For each i , let

$$S_i = \{y \in Y : H^0(Y, W_i(y)) \neq 0\}.$$

This set is closed by upper semicontinuity, applied to $\operatorname{pr}_1^* W_i \otimes \mathcal{O}_{Y \times Y}(\Delta)$ over the second projection. Since $H^0(Y, E(y)) \neq 0$ for every y , the finitely many sets S_i cover Y . The curve Y is irreducible, so one of them equals Y . Fix such an index and write $W = W_i$. Then

$$H^0(Y, W(y)) \neq 0 \quad \text{for every } y \in Y.$$

By Lemma 4.1, $f_*\mathcal{O}_X \simeq \mathcal{E}(\operatorname{Fun}(G, \mathbb{C}))$. The regular representation $\operatorname{Fun}(G, \mathbb{C})$ contains both the trivial representation and V_i as direct summands. Hence $f_*\mathcal{O}_X$ contains $\mathcal{O}_Y \oplus W$ as a direct summand. Finally, $\mathcal{O}_X(f^{-1}(y)) \simeq f^*\mathcal{O}_Y(y)$, so the projection formula gives

$$H^0(X, \mathcal{O}_X(f^{-1}(y))) \simeq H^0(Y, f_*\mathcal{O}_X \otimes \mathcal{O}_Y(y)).$$

The direct summand $\mathcal{O}_Y \oplus W$ therefore gives an injection

$$H^0(Y, \mathcal{O}_Y(y)) \oplus H^0(Y, W(y)) \hookrightarrow H^0(X, \mathcal{O}_X(f^{-1}(y))).$$

The first summand contains the canonical section of the effective divisor y , and the second is nonzero by construction. Thus the target has dimension at least two for every $y \in Y$. $\square$

ACKNOWLEDGEMENTS

The author thanks Daniel Litt for his guidance, enthusiasm, and expertise.

The author acknowledges the use of OpenAI's ChatGPT, specifically GPT-5.5 Pro for the original partial proof and GPT-5.6-sol Pro for subsequent simplification, cross-auditing, and improvement of the exposition. The central construction, proof strategy, exploratory work, and substantial portions of the initial drafting were generated autonomously by these models. The author has repeatedly revised and audited the manuscript and takes full responsibility for the accuracy of its final contents.

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